

Identifying Data Requirements for an Online Shopping System

TASK 2

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Primary Key

A **Primary Key (PK)** is a field that uniquely identifies each record in a database table. It ensures that no two records have the same value and that each record can be accessed without confusion. A primary key cannot contain duplicate or NULL values. It also helps establish relationships between different tables in the database.

Example:

If the **Customer_ID** is **C101**, no other customer can have the same Customer_ID. Similarly, every product will have a unique **Product_ID**.

Core Entities and Their Attributes

1. Customer

Stores customer information required for shopping.

Attributes:

- Customer_ID (Primary Key)
- Customer_Name
- Email
- Phone_Number
- Password
- Address
- City
- State
- Pincode

2. Category

Stores different categories of products.

Attributes:

- Category_ID (Primary Key)
 - Category_Name
 - Category_Description
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3. Product

Stores details about products available for sale.

Attributes:

- Product_ID (Primary Key)
 - Product_Name
 - Category_ID
 - Supplier_ID
 - Brand
 - Price
 - Stock
 - Product_Description
 - Product_Image
-

4. Supplier

Stores supplier information.

Attributes:

- Supplier_ID (Primary Key)
- Supplier_Name

- Contact_Number
 - Email
 - Address
-

5. Shopping Cart

Stores products selected by customers before placing an order.

Attributes:

- Cart_ID (Primary Key)
 - Customer_ID
 - Product_ID
 - Quantity
 - Total_Price
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6. Order

Stores customer order details.

Attributes:

- Order_ID (Primary Key)
 - Customer_ID
 - Order_Date
 - Total_Amount
 - Order_Status
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7. Payment

Stores payment information.

Attributes:

- Payment_ID (Primary Key)

- Order_ID
 - Payment_Method
 - Payment_Status
 - Payment_Date
 - Amount
-

8. Shipment

Stores delivery details.

Attributes:

- Shipment_ID (Primary Key)
 - Order_ID
 - Tracking_Number
 - Courier_Name
 - Shipment_Status
 - Delivery_Date
-

9. Review

Stores customer feedback.

Attributes:

- Review_ID (Primary Key)
 - Customer_ID
 - Product_ID
 - Rating
 - Review_Comment
-

Additional Features and Data Requirements

Search

Allows customers to search for products.

Data Stored:

- Search_ID
 - Search_Keyword
 - Search_Date
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Sort

Helps arrange products based on user preference.

Sort Options:

- Price: Low to High
 - Price: High to Low
 - Newest First
 - Best Selling
 - Customer Rating
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Filter

Allows users to narrow product selection.

Filter Options:

- Brand
- Price Range
- Category
- Rating
- Availability
- Color
- Size

- Gender
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Gender

Some products are displayed based on gender.

Values:

- Men
 - Women
 - Kids
 - Unisex
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Wishlist

Stores products that customers want to purchase later.

Attributes:

- Wishlist_ID
 - Customer_ID
 - Product_ID
 - Date_Added
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Conclusion

The Online Shopping System stores information related to customers, products, suppliers, categories, shopping carts, orders, payments, shipments, reviews, and additional features like search, sorting, filtering, gender selection, and wishlist. Each entity has a unique **Primary Key**, which helps maintain data integrity and build relationships between different tables in the database.